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THERMAL MANAGEMENT SYSTEM

Abstract

The thermal management system includes an HT radiator (first radiator), an LT radiator (second radiator), a nine-way valve (switching device) and a five-way valve (switching device) for switching a flow path of the heat medium, and an ECU (control device). ECU controls the nine-way valve and the five-way valve to switch between a first heat medium circuit in which HT radiator and LT radiator are separated from each other and a second heat medium circuit including a series circuit in which HT radiator and LT radiator are connected in series.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-037223 filed on Mar. 11, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a thermal management system.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2020-185829 (JP 2020-185829 A) discloses an electrified vehicle that includes an in-vehicle temperature control device including a high-temperature radiator and a low-temperature radiator. The high-temperature radiator is provided in a high-temperature circuit. The low-temperature radiator is provided in a low-temperature circuit. The high temperature circuit and the low temperature circuit are independent of each other.

SUMMARY

[0004] In JP 2020-185829 A, as described above, the high-temperature circuit and the low-temperature circuit are independent of each other. That is, the high-temperature radiator and the low-temperature radiator are provided to cool different heat sources. Therefore, there is a possibility that the cooling power is insufficient with only one radiator when the amount of heat of a heat source is large.

[0005] The present disclosure has been made to address the above issue, and has an object to provide a thermal management system capable of efficiently cooling a heat source using two radiators.

[0006] An aspect of the present disclosure provides a thermal management system in which a heat medium circulates, and the thermal management system includes: a first radiator; [0007] a second radiator that is different from the first radiator; [0008] a switching device that switches a flow path of the heat medium; and [0009] a control device that controls the switching device.

[0010] The control device controls the switching device so as to switch between [0011] a first heat medium circuit in which the first radiator and the second radiator are separated to be independent of each other, and [0012] a second heat medium circuit that includes a series circuit in which the first radiator and the second radiator are connected in series with each other.

[0013] In the thermal management system according to the aspect of the present disclosure, as described above, a second heat medium circuit that includes a series circuit in which the first radiator and the second radiator are connected in series with each other is formed. Thus, a common heat source can be cooled by both the first radiator and the second radiator. As a result, a heat source can be efficiently cooled by forming the second heat medium circuit when the amount of heat generated by the heat source is relatively large or the like. When the amount of heat generated by a heat source is relatively small, meanwhile, the cooling by the first radiator and the cooling by the second radiator can be individually performed by forming the first heat medium circuit. As a result, a heat source can be efficiently cooled when the amount of heat generated by the heat source is relatively small or the like.

[0014] The thermal management system may further include: [0015] a power storage device; and [0016] a drive device capable of generating a drive force.

[0017] In the first heat medium circuit, [0018] the first radiator may discharge heat supplied from the power storage device to outside air, and [0019] the heat medium flowing through the second radiator may perform heat exchange with the drive device.

[0020] In the second heat medium circuit, each of the first radiator and the second radiator may

discharge the heat supplied from the power storage device to the outside air.

[0021] With such a configuration, the first radiator provided for the power storage device and the second radiator provided for the drive device in the first heat medium circuit can be used in combination to cool the power storage device in the second heat medium circuit. Thus, the power storage device can be efficiently cooled by the first radiator and the second radiator in the second heat medium circuit.

[0022] The control device may form the second heat medium circuit by controlling the switching device when the power storage device is externally charged. With such a configuration, the power storage device can be efficiently cooled using both the first radiator and the second radiator during external charging in which the temperature of the power storage device is relatively likely to increase.

[0023] The control device may form the second heat medium circuit by controlling the switching device when the power storage device is externally charged through rapid charging. With such a configuration, the power storage device can be efficiently cooled using both the first radiator and the second radiator during external charging in which the temperature of the power storage device is more likely to increase than during normal charging.

[0024] The switching device may include a nine-way valve and a five-way valve. The control device may switch between the first heat medium circuit and the second heat medium circuit by controlling each of the nine-way valve and the five-way valve. With such a configuration, it is possible to easily switch between the first heat medium circuit and the second heat medium circuit by switching the flow path of the heat medium using two different multi-way valves (nine-way valve and five-way valve).

[0025] According to the present disclosure, it is possible to efficiently cool a heat source using two different radiators.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0027] FIG. 1 is a diagram illustrating a configuration of a vehicle equipped with a thermal management system according to an embodiment;

[0028] FIG. 2 is a diagram illustrating a configuration of a thermal management circuit of a thermal management system according to an embodiment;

[0029] FIG. 3 is a diagram illustrating a first heat medium circuit of a heat management circuit according to an embodiment;

[0030] FIG. 4 is a diagram illustrating a second heat medium circuit of a heat management circuit according to an embodiment;

[0031] FIG. 5 is a diagram illustrating a third heat medium circuit of a heat managing circuit according to an embodiment; and

[0032] FIG. 6 is a diagram illustrating a control flowchart of a ECU according to an embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0033] Hereinafter, an embodiment of the present disclosure will be described in detail with reference to the drawings. The same or corresponding parts in the drawings are designated by the same reference characters and repetitive description will be omitted.

[0034] As illustrated in FIG. 1, a configuration in which the thermal management system **10** according to the present disclosure is mounted on a electrified vehicle **20** will be described below. Electrified vehicle **20** is preferably a vehicle equipped with a battery **510** (described below, FIG. 2)

for traveling. Electrified vehicle **20** is, for example, battery electric vehicle (BEV: Battery Electric Vehicle). Alternatively, electrified vehicle **20** is, for example, hybrid electric vehicle (HEV: Hybrid Electric Vehicle), plug-in hybrid electric vehicle (PHEV: Plug-in Hybrid Electric Vehicle), or fuel cell electric vehicle (FCEV: Fuel Cell Electric Vehicle). However, the use of the thermal management system according to the present disclosure is not limited to a vehicle.

[0035] The thermal management system **10** includes a thermal management circuit **1** and an ECU (Electronic Control Unit) **2**. Note that ECU **2** is an exemplary “control device” of the present disclosure.

[0036] ECU **2** includes a processor **2a**, a memory **2b**, a storage **2c**, and an interface **2d**.

[0037] The processor **2a** is, for example, CPU (Central Processing Unit) or MPU (Micro-Processing Unit). The memory **2b** is, for example, RAM (Random Access Memory). The storage **2c** is a rewritable non-volatile memory such as HDD (Hard Disk Drive), SSD (Solid State Drive), and flash memory. The storage **2c** stores a system program including OS (Operating System) and a control program including computer-readable code required for control operations. The processor **2a** realizes various processes by reading out a system program and a control program, expanding the program into a memory **2b**, and executing the program. The interface **2d** controls the communication between ECU **2** and the components of the thermal management circuit **1**.

[0038] ECU **2** generates a control command based on a sensor value acquired from various sensors included in the thermal management circuit **1**, a user manipulation received by an HMI (Human Machine Interface) (not shown) provided in electrified vehicle **20**, and the like. ECU **2** outputs the generated control command to the thermal management circuit **1**. ECU **2** may be divided into a plurality of ECU for each function. Further, although FIG. **1** shows an exemplary ECU **2** including one processor **2a**, ECU **2** may include a plurality of processors. The same applies to the memory **2b** and the storage **2c**.

Configuration of Thermal Management Circuit

[0039] FIG. **2** is a diagram illustrating an example of a configuration of the thermal management circuit **1** according to the present embodiment. The thermal management circuit **1** includes a nine-way valve **100**, a five-way valve **200**, a high-temperature circuit **300**, a unit circuit **400**, a battery circuit **500**, a refrigeration cycle **600**, a flow path **700**, and a flow path **800**. The thermal management circuit **1** includes a flow path **30**, a flow path **40**, a flow path **50**, and a flow path **60**. Note that each of the nine-way valve **100** and the five-way valve **200** is an example of a “switching device” of the present disclosure.

[0040] The nine-way valve **100** includes a valve body **110** and outer sections **120** to **129**. The valve body **110** has a cylindrical shape extending in the Z direction, and is configured to be rotatable about a central axis line (not shown) of the valve body **110**. The valve body **110** is surrounded by the outer sections **120** to **129** when viewed from Z1. The valve body **110** rotates in accordance with a control command from ECU **2** (FIG. **1**).

[0041] The valve body **110** is provided with an internal flow path **111**, an internal flow path **112**, an internal flow path **113**, and an internal flow path **114**. When the valve body **110** rotates in accordance with a control command from ECU **2**, the positions of the internal flow paths **111** to **114** are changed. As a result, the connection (connection combination) between the outer sections **120** to **129** and the internal flow paths **111** to **114** is changed. Details will be described later.

[0042] The internal flow path **111** is disposed Z1 the internal flow path **112**. The internal flow path **113** is disposed Z1 the internal flow path **114**. The internal flow path **111** and the internal flow path **113** are arranged at the same position in the Z direction. The internal flow path **112** and the internal flow path **114** are arranged at the same position in the Z direction.

[0043] The outer sections **120** to **129** are circumferentially arranged on the outer peripheral side of the valve body **110**. The outer sections **120** to **129** are isolated from each other. Although the outer sections **128** and **129** are shown as being arranged side by side in FIG. **2** for clarity, they are actually stacked in the Z direction. The outer section **129** is disposed Z2 the outer section **128**.

[0044] Each of the outer sections **120** to **126** has substantially the same width in the circumferential direction of the valve body **110**. The circumferential width of each of the outer sections **127** to **129** is less than the circumferential width of the outer sections **120** to **126**.

[0045] The five-way valve **200** has a cylindrical shape extending in the Z direction. The five-way valve **200** has ports **P1** to **P5**. **P4** from the port **P1** is an inlet port that allows the heating medium to flow into the five-way valve **200**. The port **P5** is an outlet port that allows the heating medium to flow out of the five-way valve **200**. **P4** from the port **P1** is in communication with a lower compartment (not shown) that is isolated from one another. That is, the five-way valve **200** is provided with four lower compartments that are isolated from each other. Although not shown, the four sections have the same shape (a sector shape having a central angle of 90 degrees when viewed from **Z1** side). The port **P5** communicates with an upper compartment (not shown) provided **Z1** of the four lower compartments.

[0046] The flow condition of the heat medium from the ports **P1** to **P5** is controlled by a control command from ECU **2** (FIG. **1**). Specifically, ECU **2** changes the position of the opening **210**. Inside the five-way valve **200**, a partition plate (not shown) for partitioning the four lower and upper partitions is provided, the opening **210** is formed in the partition plate. The partition plate is rotationally moved about the central axial line of the five-way valve **200** having a cylindrical shape by a control command from ECU **2**. As a result, the lower section overlapping with the opening **210** in the Z direction changes. The port **P5** is only in communication with a port that communicates with the lower compartment that overlaps the opening **210** in the Z-direction.

[0047] The opening **210** has a sector shape with a central angle of about 90 degrees when viewed from **Z1**. Thus, one or two of the four lower compartments may simultaneously overlap the opening **210** in the Z direction. This causes one or two ports of the ports **P1** to **P4** to communicate with the port **P5**.

[0048] The high-temperature circuit **300** includes a water pump **310**, an HVH (High Voltage Heater) **320**, a heater core **330**, and an HT (High Temperature) radiator **340**. The water pump **310** circulates the heating medium in the high-temperature circuit **300** in accordance with a control command from ECU **2** (FIG. **1**). Note that HT radiator **340** is an exemplary “first radiator” of the present disclosure.

[0049] The high-temperature circuit **300** includes a flow path **350**, a flow path **360**, and a flow path **370**. The flow path **350** connects the port **P5** and the branch point **380**. The flow path **350** is provided with a water pump **310** and an HVH **320**. The flow path of the flow path **350** between the water pump **310** and HVH **320** is also connected to a water-cooled condenser **640** of the refrigeration cycle **600**, which will be described later. That is, in the water-cooled condenser **640**, heat exchange is performed between the heat medium circulating in the refrigeration cycle **600** and the heat medium flowing through the flow path **350**.

[0050] The flow path **360** connects the branch point **380** and the port **P2**. The flow path **360** is provided with a heater core **330**.

[0051] The flow path **370** connects the branch point **380** and the port **P1**. A HT radiator **340** is provided in the flow path **370**.

[0052] unit circuit **400** includes an LT (Low Temperature) radiator **410**, a reserve tank **420**, and a water pump **430**, unit circuit **400** also includes SPU (Smart Power Unit) **440**, PCU (Power Control Unit) **450**, oil cooler (O/C) **460**, and transaxle (T/A) **470**. The water pump **430** circulates the heating medium in accordance with a control command from ECU **2** (FIG. **1**). The water pump **430** delivers a heat medium to SPU **413**. The transaxle **470** is capable of generating the driving force of electrified vehicle **20** (FIG. **1**). Note that LT radiator **410** is an exemplary “second radiator” of the present disclosure. In addition, each of PCU **450** and the transaxle **470** is an exemplary “drive device” of the present disclosure.

[0053] The unit circuit **400** includes a flow path **480** and a flow path **490**. The flow path **480** connects the outer section **127** and the outer section **120** of the nine-way valve **100**. The flow path

480 is provided with a LT radiator **410**, a reserve tank **420**, a water pump **430**, an SPU **440**, PCU **450**, and an oil cooler **460**. Each of SPU **440** and PCU **450** includes a heat exchanger (not shown) that exchanges heat with a heat medium of the flow path **480**. The oil cooler **460** exchanges heat between the transaxle **470** and the heat medium in the flow path **480**. Instead of the oil cooler **460**, a transaxle **470** may be provided in the flow path **480**.

[0054] The flow path **490** connects the outer section **129** of the nine-way valve **100** and the reserve tank **420**.

[0055] The battery circuit **500** includes a battery **510**, a water pump **520**, a flow path **530**, and a temperature sensor **540**. The water pump **520** delivers a heat medium to the battery **510** side. The water pump **520** circulates the heat medium in accordance with a control command from ECU 2 (FIG. 1). Note that the battery **510** is an example of a “power storage device” of the present disclosure.

[0056] The flow path **530** connects the outer section **123** and the outer section **124** of the nine-way valve **100**. The flow path **530** is provided with a battery **510** and a water pump **520**. The battery **510** includes a heat exchanger (not shown) that exchanges heat with the heat medium of the flow path **530**.

[0057] The temperature sensor **540** detects the temperature of the battery **510**. The data detected by the temperature sensor **540** is transmitted to ECU 2 (FIG. 1).

[0058] The refrigeration cycle **600** includes a chiller **610**, an evaporator **620**, a compressor **630**, and a water-cooled condenser **640**. The refrigeration cycle **600** includes an expansion valve **650**, an expansion valve **660**, and an EPR (Evaporative Pressure Regulator) **670**. A heat medium (a gas-phase refrigerant or a liquid-phase refrigerant) circulating in the refrigeration cycle **600** flows through one/both of the first path and the second path. The first path is the path of compressor **630**, water cooled condenser **640**, expansion valve **660**, evaporator **620**, EPR **670**, and compressor **630**. The second path is the path of compressor **630**, water cooled condenser **640**, expansion valve **650**, chiller **610**, and compressor **630**.

[0059] The flow path **700** connects the outer section **125** and the outer section **126** of the nine-way valve **100**. The flow path **700** is also connected to the chiller **610** of the refrigeration cycle **600**. That is, the heat medium flowing through the flow path **700** and the heat medium of the refrigeration cycle **600** are heat-exchanged in the chiller **610**.

[0060] The flow path **800** connects the outer section **121** and the outer section **122** of the nine-way valve **100**. A device or the like is not provided in the flow path **800**.

[0061] The flow path **30** connects the branch point **371** and the merging portion **481**. The branch point **371** is provided between HT radiator **340** and the five-way valve **200** in the flow path **370** of the high-temperature circuit **300**. The merging portion **481** is provided between LT radiator **410** and the nine-way valve **100** in the flow path **480** of the unit circuit **400**. That is, the heat medium of the flow path **370** branched into the flow path **30** at the branch point **371** is merged into the heat medium of the flow path **480** at the merging portion **481**.

[0062] The flow path **40** connects the port P3 of the nine-way valve **100** and a branch point **531** of the flow path **530** of the battery circuit **500** provided between the battery **510** and the outer section **124**. That is, the heat medium in the flow path **530** branched into the flow path **40** at the branch point **531** flows into the five-way valve **200** from the port P3.

[0063] The flow path **50** connects the branch point **380** of the high-temperature circuit **300** and the branch point **531** of the battery circuit **500**. That is, the heat medium of the high-temperature circuit **300** (the flow path **350**) branched into the flow path **50** at the branch point **380** merges into the flow path **530** at the branch point **531**.

[0064] The flow path **60** connects the port P4 of the five-way valve **200** and the branch point **482** of the flow path **480** of the unit circuit **400** between the oil cooler **460** and the outer section **120** of the nine-way valve **100**. That is, the heat medium in the flow path **480** branched into the flow path **60** at the branch point **482** flows into the five-way valve **200** from the port P4.

[0065] Here, in the conventional thermal management system, HT radiator and LT radiator cool heat sources differently from each other. Therefore, when the amount of heat of the heat source is large, the cooling power may be insufficient by only one of the radiators.

[0066] Therefore, in the present embodiment, ECU 2 controls the nine-way valve **100** and the five-way valve **200** to switch between the first heat medium circuit **1a** (FIG. 3) and the second heat medium circuit **1b** (FIG. 4). In the first heat medium circuit **1a** (FIG. 3), HT radiator **340** and LT radiator **410** are separated from each other and are independent of each other. The second heat medium circuit **1b** (FIG. 4) includes a series circuit **31** in which HT radiator **340** and LT radiator **410** are connected in series. Each of the first heat medium circuit **1a** and the second heat medium circuit **1b** is a circuit for cooling the battery **510**.

First heat Medium Circuit

[0067] FIG. 3 is a diagram illustrating a first heat medium circuit **1a** of the thermal management circuit **1**. In the first heat medium circuit **1a**, HT radiator **340** and LT radiator **410** are provided in separate circulation circuits. Discuss in detail below

[0068] The internal flow path **111** of the nine-way valve **100** connects the outer section **121** and the outer section **127**. The internal flow path **112** connects the outer section **120** and the outer section **122**. The internal flow path **113** connects the outer section **124** and the outer section **126**. The internal flow path **114** connects the outer section **123** and the outer section **125**.

[0069] In the five-way valve **200**, the opening **210** overlaps only the lower compartment that communicates with the port P1. Thus, the port P5 is in communication with only the port P1 of the ports P1 to P4.

[0070] As a result, the heat medium circulates through the circuit of the five-way valve **200** (port P5), the water pump **310**, the water-cooled condenser **640**, HVH **320**, HT radiator **340**, and the five-way valve **200** (port P1) (the circulation circuit in the high-temperature circuit **300**).

[0071] The heating medium also circulates in the circuitry of the reserve tank **420**, water pump **430**, SPU **440**, PCU **450**, oil cooler **460**, outer compartment **120**, inner channel **112**, outer compartment **122**, channel **800**, outer compartment **121**, inner channel **111**, outer compartment **127**, LT radiator **410**, and reserve tank **420**. In the first heat medium circuit **1a**, the heat medium flowing through LT radiator **410** exchanges heat with SPU **440**, PCU **450** and the transaxle **470** (oil cooler **460**).

[0072] In addition, the heat medium circulates in the circuits of the water pump **520**, the battery **510**, the outer section **124**, the internal flow path **113**, the outer section **126**, the chiller **610**, the outer section **125**, the internal flow path **114**, the outer section **123**, and the water pump **520**.

[0073] In addition, in the refrigeration cycle **600**, the heat medium circulates through the circuit of the chiller **610**, compressor **630**, water-cooled condenser **640**, expansion valve **650**, and chiller **610**.

[0074] In the first heat medium circuit **1a**, HT radiator **340** discharges heat (heat generated in the battery **510**) supplied from the battery **510** to the outside air. The heat of the battery **510** is transferred to the high-temperature circuit **300** through the chiller **610** (refrigeration cycle **600**) and the water-cooled condenser **640**, and is discharged from HT radiator **340** to the outside air.

[0075] In the first heat medium circuit **1a**, the heat medium does not flow through each of the flow path **30**, the flow path **40**, the flow path **50**, and the flow path **60**. In the first heat medium circuit **1a**, the heat medium does not flow through the flow path **490**.

Second Heat Medium Circuit

[0076] FIG. 4 is a diagram illustrating a second heat medium circuit **1b** of the thermal management circuit **1**. In the second heat medium circuit **1b**, HT radiator **340** and

[0077] LT radiator **410** are provided in a common circulation circuit.

[0078] The internal flow path **111** of the nine-way valve **100** connects the outer section **121** and the outer section **128**. The internal flow path **112** connects the outer section **120** and the outer section **122**. The internal flow path **113** connects the outer section **124** and the outer section **126**. The internal flow path **114** connects the outer section **123** and the outer section **125**.

[0079] In the five-way valve **200**, the opening **210** overlaps only the lower compartment that

communicates with the port P4. Thus, the port P5 is in communication with only the port P4 of the ports P1 to P4.

[0080] As a result, the heating medium circulates through the circuitry of the five-way valve 200 (port P5), the water pump 310, the water-cooled condenser 640, HVH 320, HT radiator 340, the branch point 371, the flow path 30, the merging portion 481, LT radiator 410, the reserve tank 420, the water pump 430, SPU 440, PCU 450, oil cooler 460, the branch point 482, the flow path 60, and the five-way valve 200 (port P4). A series circuit 31 in which HT radiator 340 and LT radiator 410 are connected in series is formed in the circulation circuit.

[0081] In addition, the heat medium circulates in the circuits of the water pump 520, the battery 510, the outer section 124, the internal flow path 113, the outer section 126, the chiller 610, the outer section 125, the internal flow path 114, the outer section 123, and the water pump 520.

[0082] In addition, in the refrigeration cycle 600, the heat medium circulates through the circuit of the chiller 610, compressor 630, water-cooled condenser 640, expansion valve 650, and -chiller 610.

[0083] The heat of the battery 510 is transmitted to the high-temperature circuit 300 through the chiller 610 (refrigeration cycle 600) and the water-cooled condenser 640, and is transmitted to the unit circuit 400 through the flow path 30. Thus, in the second heat medium circuit 1b, each of HT radiator 340 and LT radiator 410 discharges the heat (the heat generated in the battery 510) supplied from the battery 510 to the outside air.

[0084] In the second heat medium circuit 1b, the heat medium does not flow through each of the flow path 50 and the flow path 60. In the second heat medium circuit 1b, the heat medium does not flow through the flow path 490.

Third Heat Medium Circuit

[0085] FIG. 5 is a diagram illustrating a third heat medium circuit 1c of the thermal management circuit 1. The third heat medium circuit 1c is a circuit for passing water to the respective devices of the thermal management circuit 1. Thus, it is possible to eliminate the liquid accumulation and the like.

[0086] The internal flow path 111 of the nine-way valve 100 connects the outer section 122 and the outer section 124. The internal flow path 112 connects the outer section 123 and the outer section 125. The internal flow path 113 connects the outer section 121 and the outer section 127. The internal flow path 114 connects the outer section 120 and the outer section 126.

[0087] In the five-way valve 200, the opening 210 overlaps the lower compartment that communicates with each of the ports P1 and P2. Thus, the port P5 is in communication with the ports P1 and P2 of the ports P1 to P4.

[0088] Thus, the heating medium circulates through the first circuitry of the five-way valve 200 (port P5)-water pump 310-water cooled condenser 640-HVH 320-heater core 330-five-way valve 200 (port P2).

[0089] The heating medium also circulates through the second circuit of the five-way valve 200 (port P5), water pump 310, water cooled condenser 640, HVH 320, HT radiator 340, and five-way valve 200 (port P1).

[0090] The heat medium of the second circuit is branched into the flow path 30 at the branch point 371. The heat medium branched into the flow path 30 circulates through a third circuit of LT radiator 410, reserve tank 420, water pump 430, SPU 440, PCU 450, oil cooler 460, outer section 120, internal flow path 114, outer section 126, chiller 610, outer section 125, internal flow path 112, outer section 123, water pump 520, battery 510, outer section 124, inner flow path 111, outer section 122, flow path 800, outer section 121, internal flow path 113, outer section 127, and LT radiator 410.

[0091] The heat medium of the third circuit is branched into the flow path 50 at the branch point 531, and flows into the high-temperature circuit 300 from the branch point 380.

[0092] In the third heat medium circuit 1c, the series circuit 31 is formed similarly to the second

heat medium circuit **1b**. In the third heat medium circuit **1c**, the heat medium does not flow through each of the flow path **40** and the flow path **60**. In the third heat medium circuit **1c**, the heat medium does not flow through the flow path **490**.

ECU Control Flowchart

[0093] FIG. **6** is a diagram illustrating an exemplary control flow executed by ECU **2** (FIG. **1**). The control flow illustrated in FIG. **6** may be executed (started) at a predetermined cycle (for example, every minute).

[0094] In **S1**, ECU **2** determines whether the battery-cooling flag is ON. The cell-cooling flag is turned ON when the detected value of the temperature sensor **540** that detects the temperature of the battery **510** is equal to or higher than a predetermined value (for example, 40° C.). If the cell-cooling flag is ON (Yes in **S1**), the process proceeds to **S2**. If the cell-cooling flag is OFF (No in **S1**), the process ends.

[0095] In **S2**, ECU **2** determines whether the charging method of the battery **510** is rapid charging (whether the rapid charging mode is ON). ECU **2** determines the charging method based on the information of the connector of electrified vehicle **20** to which the charging plug is connected, the information of the charging current, the information transmitted from the charging station, and the like. If the charging scheme is rapid charging (Yes in **S2**), the process proceeds to **S3**. If the charging scheme is not quick charging (normal charging) (No in **S2**), the process proceeds to **S4**.

[0096] In **S3**, ECU **2** controls the five-way valve **200** and the nine-way valve **100** to switch the thermal management circuit **1** to the second heat medium circuit **1b** (FIG. **4**). The battery **510** is then cooled by both HT radiator **340** and LT radiator **410**. When the thermal management circuit **1** is switched to the second heat medium circuit **1b**, the condition is maintained.

[0097] In **S4**, ECU **2** controls the five-way valve **200** and the nine-way valve **100** to switch the thermal management circuit **1** to the first heat medium circuit **1a** (FIG. **3**). Here, the battery **510** is cooled only by HT radiator **340** of HT radiator **340** and LT radiator **410**. When the thermal management circuit **1** is switched to the first heat medium circuit **1a**, the condition is maintained.

[0098] In addition, ECU **2** may control the five-way valve **200** and the nine-way valve **100** to switch the thermal management circuit **1** to the third heat medium circuit **1c** (FIG. **5**) while the cell-cooling flag is OFF.

[0099] As described above, in the present embodiment, ECU **2** controls the five-way valve **200** and the nine-way valve **100** to switch between the first heat medium circuit **1a** and the second heat medium circuit **1b**. In the first heat medium circuit **1a**, HT radiator **340** and LT radiator **410** are separated from each other and are independent of each other.

[0100] The second heat medium circuit **1b** includes a series circuit **31** in which HT radiator **340** and LT radiator **410** are connected in series. This makes it possible to easily switch between a mode in which the battery **510** is cooled by using two radiators (HT radiator **340** and LT radiator **410**) and a mode in which the battery **510** is cooled by using only one radiator (HT radiator **340**). Thus, by switching the two modes according to the temperature of the battery **510**, the battery **510** can be cooled efficiently and with an appropriate cooling power.

[0101] In the above-described embodiment, an example in which a radiator for cooling the battery **510** is switchable has been described, but the present disclosure is not limited thereto. A radiator that cools a heat source other than the battery **510** (e.g., a PCU **450** or the like) may be switchable.

[0102] In the above embodiment, the thermal management circuit **1** is switched during external charging, but the present disclosure is not limited thereto. The thermal management circuit **1** may be switched at a time other than external charging. For example, the thermal management circuit **1** may be switched during high-load driving of the unit circuit **400** or the like. In this case, the switching control for connecting HT radiator **340** and LT radiator **410** in series may be performed only when the temperature of the heat medium of the unit circuit **400** is equal to or lower than a predetermined value (for example, 50° C.). As a result, it is possible to suppress the temperature of the unit circuit **400** becoming excessively high (for example, becoming 65° C. or higher).

[0103] In the above-described embodiment, the second heat medium circuit **1b** is formed during external charging by rapid charging. For example, the second heat medium circuit **1b** may be formed during external charging by normal charging.

[0104] In the above embodiment, the state of the thermal management circuit **1** is switched by controlling the nine-way valve **100** and the five-way valve **200**, but the present disclosure is not limited thereto. The switching valve may have other configurations (e.g., a multi-way valve other than a five-way valve and a nine-way valve).

[0105] In the above embodiment, PCU **450** and the oil cooler **460** (transaxle **470**) are connected in series, but the present disclosure is not limited thereto. PCU **450** and the oil cooler **460** (transaxle **470**) may be connected in parallel.

[0106] Note that the configurations (processes) of the above-described embodiments and the above-described modification examples may be combined with each other.

[0107] The embodiment disclosed herein shall be construed as exemplary and not restrictive in all respects. The scope of the present disclosure is shown by the claims rather than by the above description of the embodiments, and is intended to include all modifications within the meaning and scope equivalent to those of the claims.

Claims

1. A thermal management system in which a heat medium circulates, the thermal management system comprising: a first radiator; a second radiator that is different from the first radiator; a switching device that switches a flow path of the heat medium; and a control device that controls the switching device, wherein the control device controls the switching device so as to switch between a first heat medium circuit in which the first radiator and the second radiator are separated to be independent of each other, and a second heat medium circuit that includes a series circuit in which the first radiator and the second radiator are connected in series with each other.
 2. The thermal management system according to claim 1, further comprising: a power storage device; and a drive device capable of generating a drive force, wherein in the first heat medium circuit, the first radiator discharges heat supplied from the power storage device to outside air; the heat medium flowing through the second radiator performs heat exchange with the drive device, and in the second heat medium circuit, each of the first radiator and the second radiator discharges the heat supplied from the power storage device to the outside air.
 3. The thermal management system according to claim 2, wherein the control device forms the second heat medium circuit by controlling the switching device when the power storage device is externally charged.
 4. The thermal management system according to claim 3, wherein the control device forms the second heat medium circuit by controlling the switching device when the power storage device is externally charged through rapid charging.
 5. The thermal management system according to claim 1, wherein: the switching device includes a nine-way valve and a five-way valve; and the control device switches between the first heat medium circuit and the second heat medium circuit by controlling each of the nine-way valve and the five-way valve.
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